

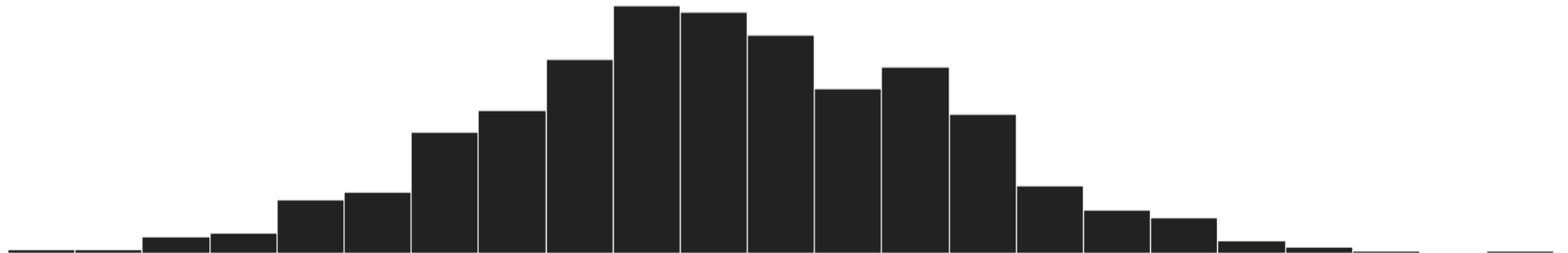
VTPEH 6105

Lecture 08

COMPARING WHAT IS COMPARABLE...

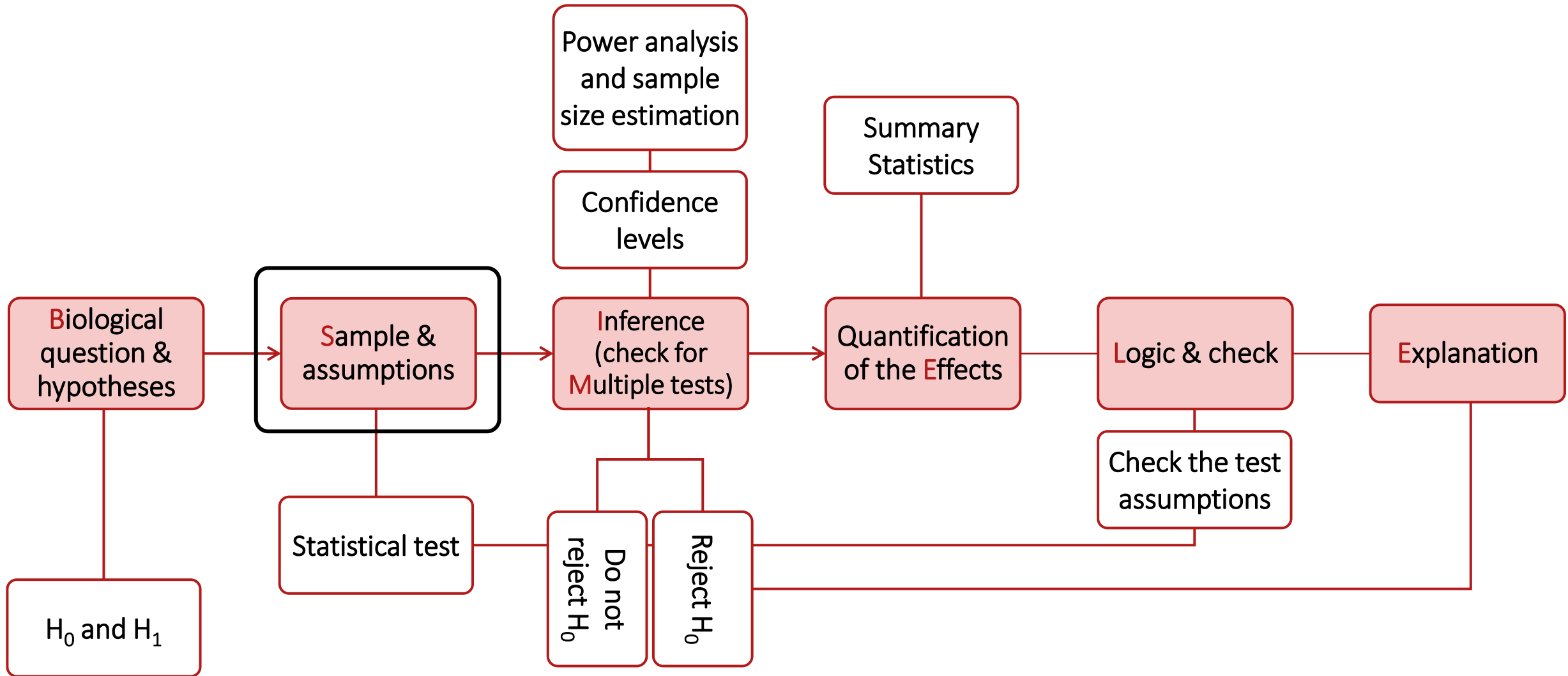
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House Keeping

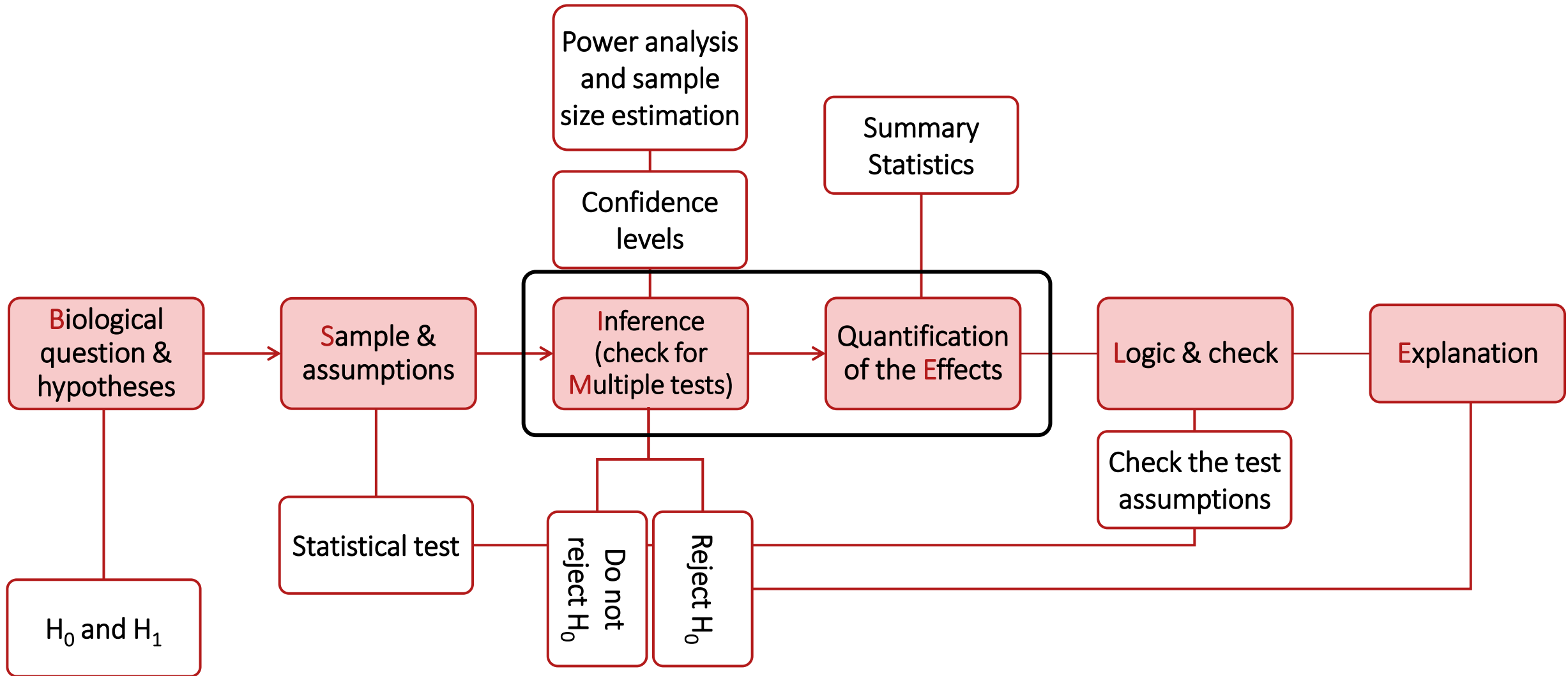
Feedback on Previous Homework: Key Missing Parts



Feedback on Previous Homework: Key Missing Parts

- **You need to check the test assumptions!**
- Continuous variable ~ categorical variable
 - T-test
 - Wilcoxon
 - ANOVA
 - Kruskal-Wallis
- Categorical variable ~ categorical variable
 - Chi2
 - Fischer
- Continuous variable ~ continuous variable
 - Pearson
 - Spearman
 - Linear regression model

Feedback on Previous Homework: Key Missing Parts



Feedback on Previous Homework: Key Missing Parts

- You need to say exactly what differs from what, using post-hoc test if necessary (and how much it differs)
- Continuous variable ~ categorical variable
 - T-test
 - Wilcoxon
 - ANOVA
 - Kruskal-Wallis
- Categorical variable ~ categorical variable
 - Chi2
 - Fischer
- Continuous variable ~ continuous variable
 - Pearson
 - Spearman
 - Linear regression model

Before Next Time (Friday March 20, in LH4)

- Practice: **Homework #08** due by **Thursday 5 PM**
- Use the Guiding Quiz #08 if needed
- Check you are ready for next class: **Knowledge Check #08** due by **Friday 11 AM**

Need Help?

- Come to VTPEH 6108: Public Health Data Lab
- Come to office hours (in-person or zoom)
 - **Tuesdays, 09:00-10:00 AM**
 - **Wednesdays, 05:00-06:00 PM**
 - If you cannot make it at these times, please contact the course assistant of your choice by email
 - In the **Old MPH Writing Room**, unless indicated otherwise on Canvas. Zoom link on syllabus



RATES AND STANDARDIZATION

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March 2026



Learning objectives

- Understand and define concepts:
 - counts, proportions, rates, ratios,
 - crude rates, age-specific rates,
 - standardization
 - standard population, population of interest,
 - Stratum adjusted /age-adjusted rates.
- Understand the use of standardization in public health.
- Understand, describe and calculate standardized rates
 - Direct method
 - Indirect method
- Discuss whether directly or indirectly standardized rates of two populations can be compared
- Discuss the strengths and limitations of direct and indirect standardization.



Counts

Number of observations of a particular event.

The numbers represent actual measurable quantities.

Can be used to convey the magnitude of a problem.

- More deaths (counts) due to unintentional injuries in 2021
- fewer deaths (counts) due to chronic lower respiratory diseases in 2021
- fewer deaths (counts) due to Alzheimer's disease in 2021



Ratio

Is a fraction where the numerator is not a part of the denominator.

A ratio is the relative magnitude of two quantities or a comparison(counts) of any two values.

Example: person-doctor ratio, person-hospital ratio, male-female ratio.

$$\text{Ratio} = \frac{c}{d} * k$$

k= constant used in ratio is 1



Ratio

	N	Deaths
Diabetic men	189	100
Nondiabetic men	3,151	811
Diabetic women	218	72
Nondiabetic women	3,823	511

- Calculate the ratio of nondiabetic women to nondiabetic men.

- $\text{Ratio} = 3823 / 3151 = 1.21 \times 1 = 1.21$

- The number of nondiabetic women is 1.21 times higher than that of men.



Proportions

- Proportions are measures that state a count relative to the size of the group.
- A proportion is the comparison of a part to the whole.
- A proportion may be expressed as a decimal, a fraction, or a **percentage**.
- Proportion = $a / (a + b) * k$
- a = MPH students younger than 25 years.
- $a + b$ = Total number of MPH students.
- k usually 100

Example:

- You might use a proportion to describe what percentage of MPH students are younger than 25 years.



Proportions

	N	Deaths
Diabetic men	189	100
Nondiabetic men	3,151	811
Diabetic women	218	72
Nondiabetic women	3,823	511

- Calculate the proportion of deaths among men.

Numerator = deaths in diabetic men = 100

Deaths in diabetic men + 811 deaths in nondiabetic men = **911 deaths in men**

Denominator = all deaths = 911 deaths in men + 72 deaths in diabetic women + 511 deaths in nondiabetic women = 1,494 deaths

Proportion = $911 / 1,494 = 0.609 \times 100 \approx 61\%$

Notice that the numerator (911 deaths in men) is a subset of the denominator.



Rates [Recap Epi - VTPEH6014]

Rates?

Number of events

Period of time

Population at that time

- Measures the frequency with which an event occurs in a defined population over a specified period of time.
- A rate does incorporate units of measurement and is dependent upon a measure of time.

Example of rate:

A death rate is the number of deaths occurring in a population over a period of time.



Commonly Used Measures of Disease Frequency in Public Health

- Morbidity rate
- Crude mortality (or death) rate
- Age-specific mortality rate
- Age-adjusted (standardized) mortality rate
- Cause-specific mortality rate
- Infant mortality rate
- Neonatal mortality rate
- Perinatal mortality rate
- Race-adjusted infant mortality rate
- Maternal mortality rate
- Case fatality rate
- Cancer mortality rate
- Livebirth rate
- Birth defect rate
- Attack rate
- Survival rate
- Years of potential life lost



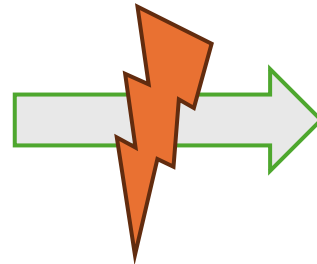
Formula Number	Name	Formula
15.1.1	Rate	$\left(\frac{a}{a+b}\right)k$
15.1.2	Ratio	$\left(\frac{c}{d}\right)k$
15.2—death rates and ratios	Annual crude death rate	$\frac{\text{Total number of deaths during year}}{\text{Total population as of July 1}} \cdot k$
	Annual specific death rate	$\frac{\text{Total number of deaths in a specific subgroup during a year}}{\text{Total population in the specific subgroup as of July 1}} \cdot k$
	Age-adjusted death rate	$\frac{\text{Total number of expected deaths}}{\text{Total standard population}} \cdot k$
	Maternal mortality rate	$\frac{\text{Deaths from all puerperal causes during a year}}{\text{Total live births during the year}} \cdot k$
	Infant mortality rate	$\frac{\text{Number of deaths under 1 year of age during a year}}{\text{Total number of live births during the year}} \cdot k$
	Neonatal mortality rate	$\frac{\text{Number of deaths under 28 days of age during a year}}{\text{Total number of live births during the year}} \cdot k$
	Fetal death rate	$\frac{\text{Total number of fetal deaths during a year}}{\text{Total number of deliveries during the year}} \cdot k$
	Fetal death ratio	$\frac{\text{Total number of fetal deaths during a year}}{\text{Total number of live births during the year}} \cdot k$
	Perinatal mortality	$\frac{(\text{Number of fetal deaths of 28 weeks or more}) + (\text{Infant deaths under 7 days})}{(\text{Number of fetal deaths of 28 weeks or more}) + (\text{Number of live births})} \cdot k$
	Cause-of-death ratio	$\frac{\text{Number of deaths due to a specific disease during a year}}{\text{Total number of deaths due to all causes during the year}} \cdot k$
	Proportional mortality ratio	$\frac{\text{Number of deaths in a particular subgroup}}{\text{Total number of deaths}} \cdot k$
15.3—measures of fertility	Crude birth rate	$\frac{\text{Total number of live births during a year}}{\text{Total population as of July 1}} \cdot k$
	General fertility rate	$\frac{\text{Total number of live births during a year}}{\text{Total number of women of childbearing age}} \cdot k$
	Age-specific fertility rate	$\frac{\text{Number of births to women of a certain age in a year}}{\text{Total number of women of the specified age}} \cdot k$
15.4—measures of morbidity	Incidence rate	$\frac{\text{Total number of new cases of a specific disease during a year}}{\text{Total population as of July 1}} \cdot k$
Prevalence rate	$\frac{\text{Total number of all diseases cases at a given point in time}}{\text{Total population at a given point in time}} \cdot k$	
Case-fatality ratio	$\frac{\text{Total number of deaths due to a disease}}{\text{Total number of cases due to the disease}} \cdot k$	
	Immaturity ratio	$\frac{\text{Total number of live births under 2500 grams during a year}}{\text{Total number of live births during the year}} \cdot k$
Secondary attack rate	$\frac{\text{Number of additional cases among contacts of a primary case within the maximum incubation period}}{\text{Total number of susceptible contacts}} \cdot k$	
Symbol Key	a, b, c, d = frequencies associated with the definition of rates and ratios k = multiplier for rates and ratios (usually 1000 or 100,000 or 1,000,000)	



Crude rates

An overall measure that **does not take explicit account** of the composition/ structure of the population is called crude.

Age,
Gender,
Education,
Religion,
SES,
Geographic,
Occupational,
Dietary,
Medical history



Formula

$$\text{Crude Rate} = \frac{(e)}{(p)} \times (k)$$

Components of the formula

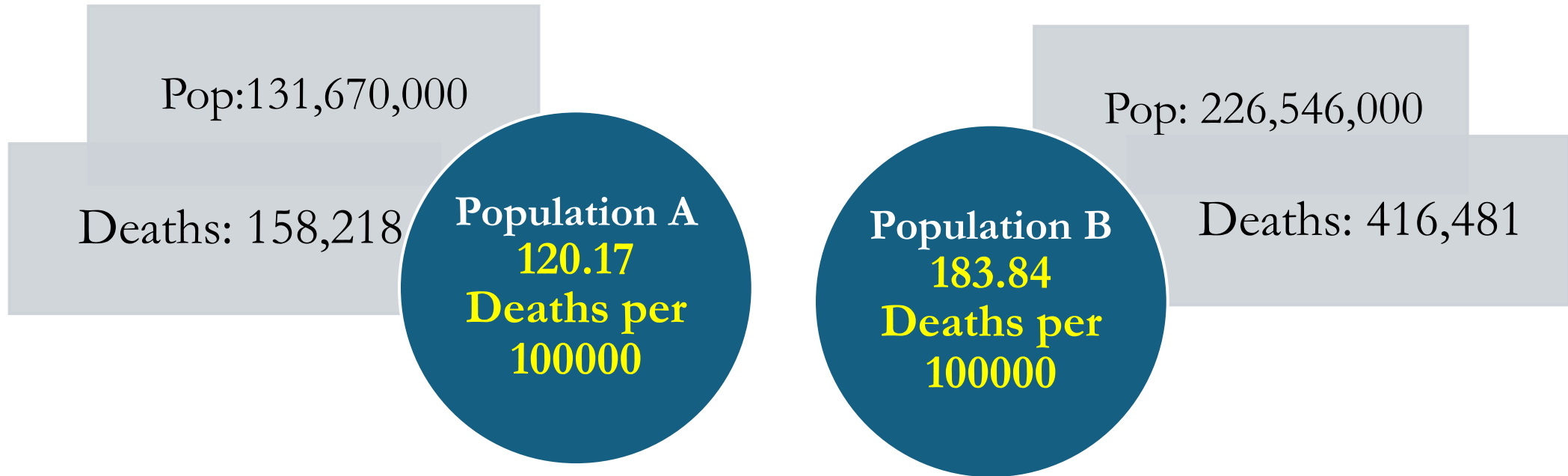
e Number of Events in the study population at a point in time

p is the study population at that point in time

k= constant a multiple of 10



Crude rates





Crudes rates

- ✓ Simple and straightforward
- ✓ Based on raw data
- ✓ Useful to provide data for allocating health resources and for public health planning.
- ✗ Does not take into account the distribution of population → subgroups or strata.
- ✗ The values may be confounded by differences between underlying population structures that can make any observed difference in crude rates difficult to interpret.
- ✗ Comparisons of crude rates between populations may not be accurate if their **demographic**/any other structures differ substantially.



Stratum-Specific Rates

Age-Specific Rates

Is the rate of an event within a specific subgroup(age-group) of the population.

Formula

$$\text{Age-specific Rate} = r_i = \frac{e_i}{p_i} \times k$$

Components of the formula

i = Stratum(age-group)

e_i = is the total observed number of events in age group i

p_i = is the population in age group i

k = constant a multiple of 10



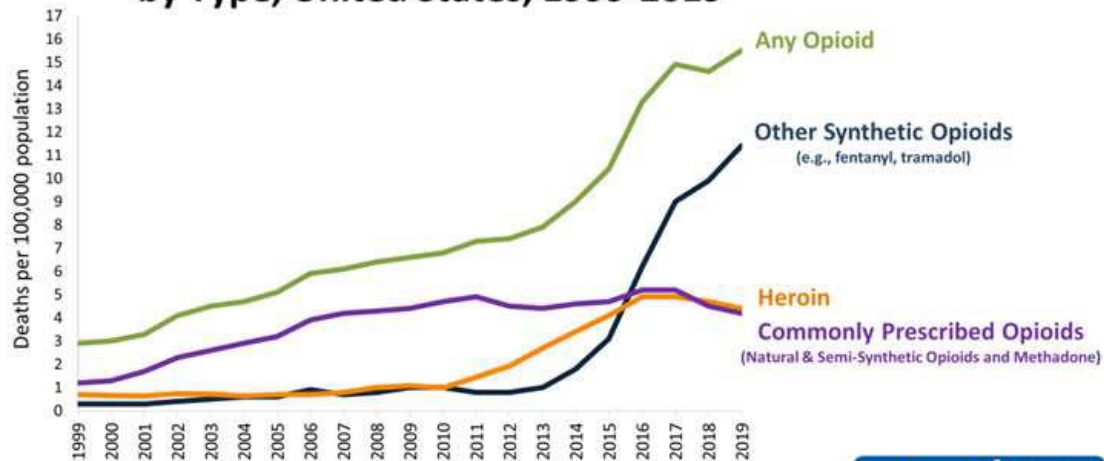
Stratum-specific rates

Population A 2020					Population B 2020		
Stratum	Age in years	Number of deaths	Population	Death rate /100000	Number of deaths	Population	Death rate /100000
1	Under 20	1,682	45,306,000	3.72	3,328	72,458,000	4.6
2	20 — 39	8,625	42,472,000	20.31	11,303	72,366,000	15.62
3	40 — 59	51,346	30,144,000	170.34	90,416	46,084,000	196.2
4	60+	96,565	13,748,000	702.4	311,434	35,638,000	873.89
Total		158,218	131,670,000	120.16	416,481	226,546,000	183.84

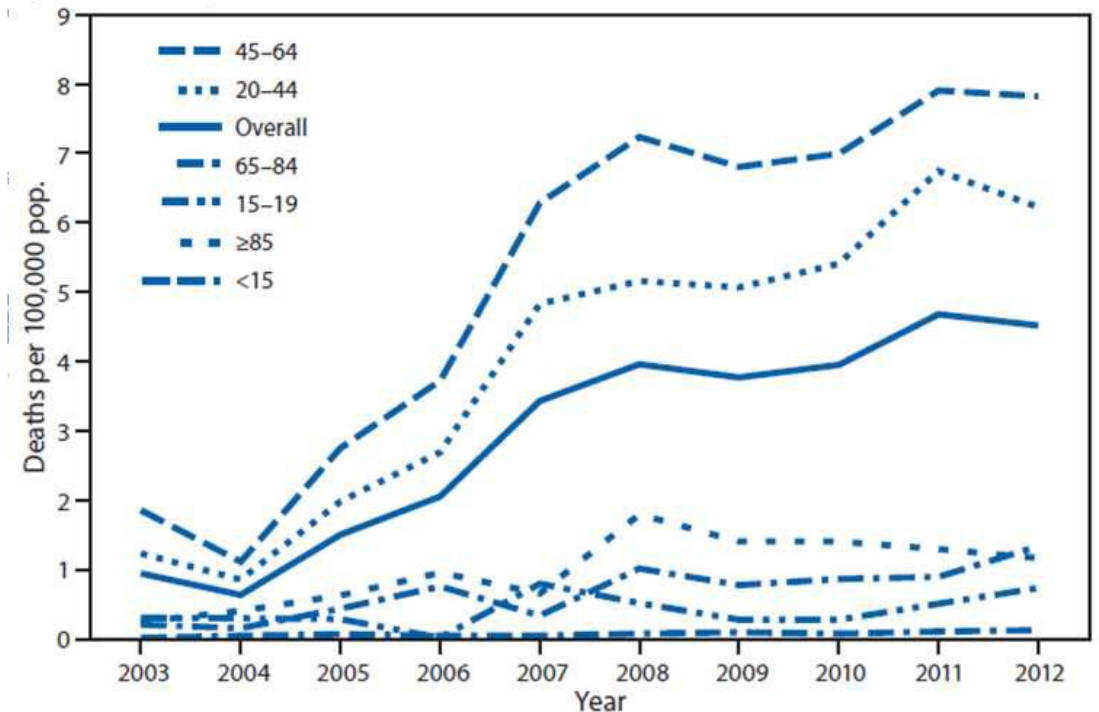


Stratum specific rates

Overdose Death Rates Involving Opioids,
by Type, United States, 1999-2019



Stratum (type)-specific rates





Stratum specific rate/ Age-Specific Rates

- ✓ More targeted
- ✓ More accurate than a comparison of the crude rates.
- ✓ Differences in the category-specific (age, sex, race) death rates may indicate different risks of death in a specific category group.
- ✓ They can be used to compare rates that may be confounded by the differences in population distribution characteristics.
- ✗ Cumbersome to compare subgroups of two or more populations.
- ✗ The calculation requires actual population data.



Standardization

- Standardization is a crucial technique in epidemiology for comparing disease rates between populations.
- Adjusts or controls for differences in population structure and provides a single summary measure for comparison across populations.
- Standardization can be used to adjust for any underlying factor (e.g., age, sex, race, SES level), or simultaneously for two or more factors in the comparison of disease rates between two or more populations.



Standardization

What we need

- Standard Population
- Population of interest (index population)
- Variable to adjust for(stratum)



Standard population and study population

- **A standard population:** Also referred to as a reference population, is a hypothetical or actual population used as a standard when calculating standardized rates.

Example: **World (WHO 2000-2025) Standard Population**

- **The study population, also referred to as the population of interest or index population,** is the actual group of individuals being observed or evaluated in a specific epidemiology study.



Examples of various common standard populations used for age standardization([*Andreas Stang et al.*](#))

Age group (years)	FRG 1987 ^{*1}	Old European Standard ^{*2}	New European Standard ^{*3}	World Standard ^{*4}	2000 U.S. Standard ^{*5}
0–4	4887	8000	6326	12 000	6914
5–9	4796	7000	6472	10 000	7253
10–14	4894	7000	6772	9000	7303
15–19	7189	7000	7208	9000	7217
20–24	8721	7000	7792	8000	6648
25–29	8044	7000	7871	8000	6453
30–34	7062	7000	7528	6000	7104
35–39	6886	7000	7212	6000	8076
40–44	6161	7000	6860	6000	8185
45–49	8043	7000	5865	6000	7212
50–54	6654	7000	5876	5000	6272
55–59	5920	6000	5553	4000	4845
60–64	5438	5000	5245	4000	3879
65–69	4338	4000	4680	3000	3426
70–74	3801	3000	2932	2000	3177
75–79	3646	2000	2897	1000	2700
80–84	2251	1000	1606	500	1784
≥ 85	1269	1000	1305	500	1551
Total	100 000	100 000	100 000	100 000	100 000

^{*1} Old states of the Federal Republic of Germany (FRG), data from the 1987 consensus

^{*2} Doll R & Cook P, 1967

^{*3} Eurostat

^{*4} Segi M, 1960

^{*5} National Cancer Institute



Standardization

- **Direct standardization**
- Indirect standardization



Direct standardization

“What would the comparable death rate be in the study populations, if they had the identical age distributions?”



Age-adjusted rate

- In the direct method, **age-specific rates(OBSERVED)** of the study population are applied to an arbitrarily chosen standard population.
- This produces the **EXPECTED** number of events, which represents the number of events that **would be expected** in the **standard population** if it **had** experienced the **same age-specific event rate** as the **study population**.



Age-adjusted rate [first method]

Step 1

- Calculate the age-specific rates of the event in the population of interest by dividing the number of cases in each age group by the total population in that age group.

Step 2

- Multiply each age-specific rate by the corresponding population in that age group from the **standard population**.

Step 3

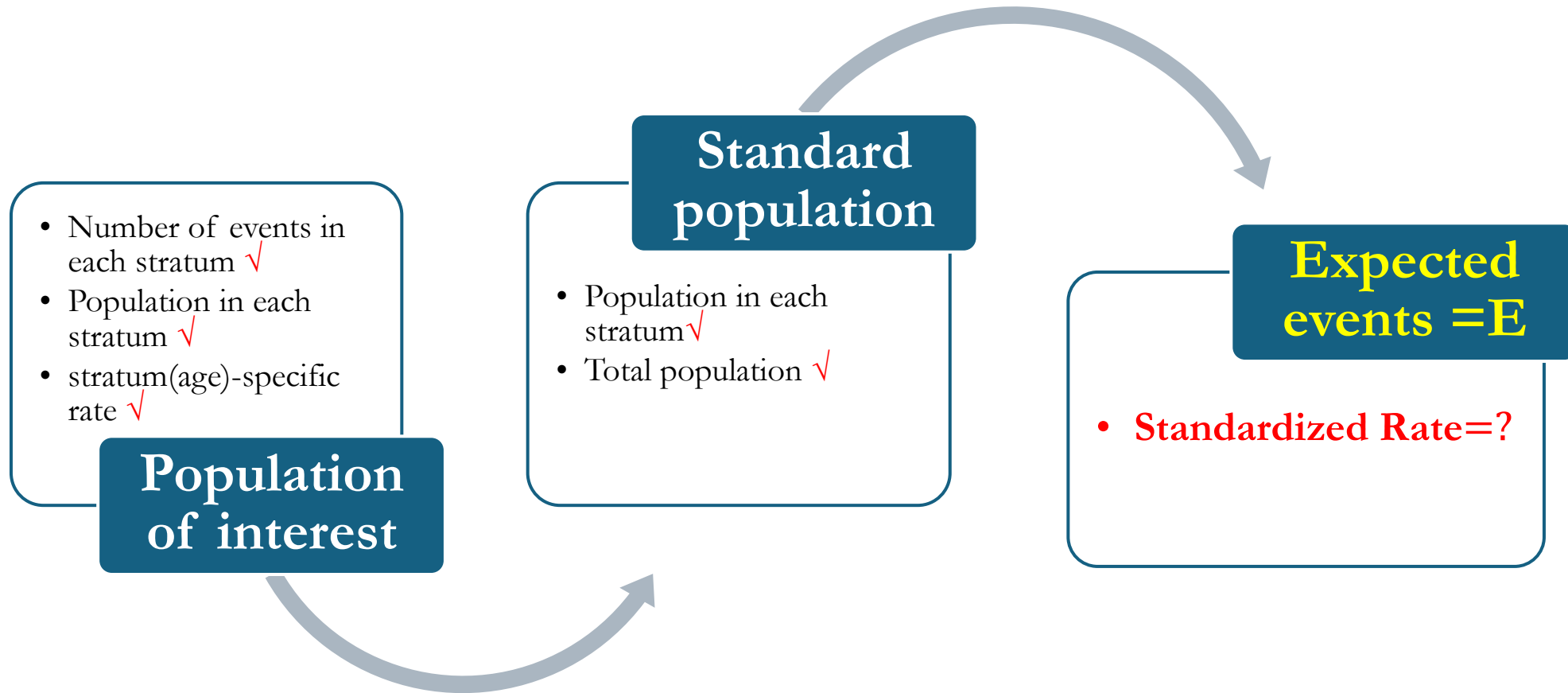
- Sum these products across all age groups to obtain the expected number of cases in each population based on the standard population's age distribution.

Step 4

- Divide the expected number of cases by the total standard population to obtain the directly standardized rate for each population. **Expressed as cases per 100,000.**



Age-adjusted rate[first method]





Age-adjusted rate: first method

Formula

$$\text{Standardized Rate} = \frac{\sum E_i}{\sum P_i} \times K$$

Components of the formula

- **E_i** is the expected number of events in age group **i** in the standard population.
- **P_i** is the Population in age group **i** in the standard population
- **K** constant
- **$E_i = \text{Age-specific expected events} = r_i * P_i$**
- **r_i** is the event rate in study population for the population in age group **i**
- **$r_i = \text{Age-specific rate} = e_i / p_i$**
- **e_i** is the observed number of events in age group **i** in the study population
- **p_i** is the population in age group **i** in the study population

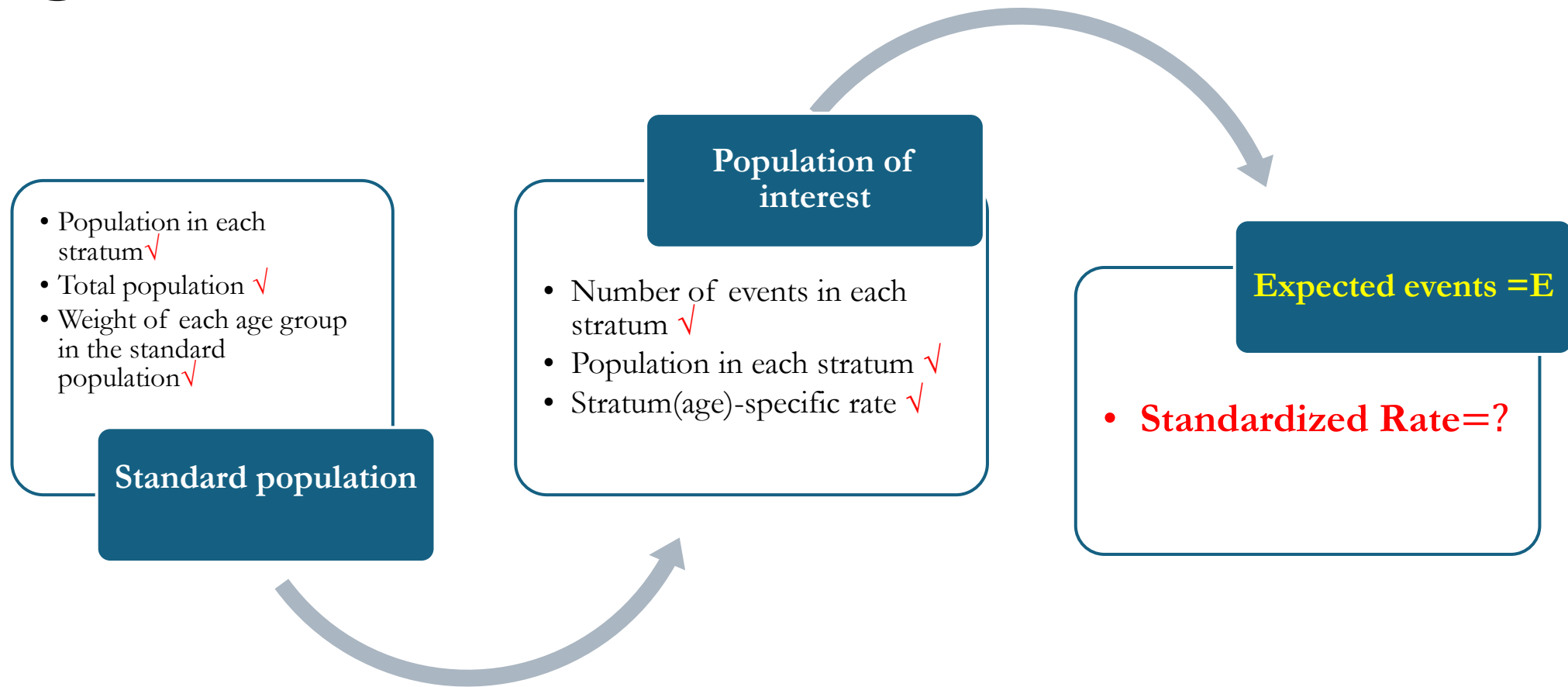


Age adjusted rate METHOD 1

Age Groups	Study Population (Ontario)		Standard Population (Canada 1991)	Age specific Rate $r_i = e/p_i$	Expected deaths $Ei = r_i * P_i$
	deaths	population			
	e_i	p_i	P_i		
<1 year	50	147,787	403,061	0.0003383	136.4
1-4	30	608,268	1,550,285	0.0000493	76.5
5-9	25	760,339	1,953,045	0.0000329	64.2
10-14	28	740,079	1,913,115	0.0000378	72.4
15 – 19	40	721,396	1,926,090	0.0000554	106.8
20 – 24	47	747,171	2,109,452	0.0000629	132.7
25 – 29	66	826,426	2,529,239	0.0000799	202.0
30 – 34	122	994,837	2,598,289	0.0001226	318.6
35 – 39	264	973,239	2,344,872	0.0002713	636.1
40 – 44	443	865,317	2,138,891	0.0005120	1095.0
45 – 49	731	790,269	1,674,153	0.0009250	1548.6
50 – 54	1,022	608,638	1,339,902	0.0016792	2249.9
55 – 59	1,523	501,218	1,238,441	0.0030386	3763.1
60 – 64	2,223	459,365	1,190,217	0.0048393	5759.8
65 – 69	3,108	431,223	1,084,588	0.0072074	7817.1
70 – 74	3,866	374,246	834,024	0.0103301	8615.6
75 – 79	3,345	255,754	622,221	0.0130790	8138.0
80 – 84	2,719	167,495	382,303	0.0162333	6206.0
85 – 89	1,692	85,585	192,410	0.0197698	3803.9
90 +	906	42,224	95,467	0.0214570	2048.4
Sum	22,250	11,100,876	28,120,065		52,791
Rate (per 100,000)				200.4346	187.7345



Age-adjusted rate[Second method]





Age-adjusted rates [second method]

Step 1

- Calculate the age-specific rates of disease of the study population by dividing the number of cases in each age group by the total population in that age group.

Step 2

- Calculate the W_i (the weight of each age group in the standard population) by dividing the population in each age group of the standard population by its total population.

Step 3

- Multiply the age-specific rate of the study population by the standard population weight to obtain the expected number of deaths.

Step 4

- Multiply the sum of these products across all age groups by a multiple a 10 ,to obtain the expected number of cases based on the standard population's age distribution. **Expressed as cases per 100,000.**



Age-adjusted mortality rates [second method]

Formula

- **Standardized Rate** = $\sum E_i * K$

Components of the formula

- **E_i** is the expected number of events in age group **i** .
- **K** constant multiple of 10
- **$E_i = \text{Age-specific expected events} = r_i * W_i$**
- **r_i** is the event rate in study population for the pop in age group **i**
- **$r_i = \text{Age-specific rate} = e_i / p_i$**
- **W_i** is the weight of each age group in the standard population.
- **$W_i = P_i / P(\text{total})$**



Age adjusted rate METHOD 2

Age Groups	Study Population (Ontario)		Standard Population (Canada 1991)	Weight	Age-specific Rate	Expected deaths
	deaths	population				
	e_i	p_i	P_i	$W_i = (P_i) / \sum (P_i)$	$r_i = e_i / p_i$	$E_i = r_i * W_i$
<1 year	50	147,787	403,061	0.014334	0.0003383	0.00000485
1-4	30	608,268	1,550,285	0.055131	0.0000493	0.00000272
5-9	25	760,339	1,953,045	0.069454	0.0000329	0.00000228
10-14	28	740,079	1,913,115	0.068034	0.0000378	0.00000257
15 - 19	40	721,396	1,926,090	0.068495	0.0000554	0.00000380
20 - 24	47	747,171	2,109,452	0.075016	0.0000629	0.00000472
25 - 29	66	826,426	2,529,239	0.089944	0.0000799	0.00000718
30 - 34	122	994,837	2,598,289	0.092400	0.0001226	0.00001133
35 - 39	264	973,239	2,344,872	0.083388	0.0002713	0.00002262
40 - 44	443	865,317	2,138,891	0.076063	0.0005120	0.00003894
45 - 49	731	790,269	1,674,153	0.059536	0.0009250	0.00005507
50 - 54	1,022	608,638	1,339,902	0.047649	0.0016792	0.00008001
55 - 59	1,523	501,218	1,238,441	0.044041	0.0030386	0.00013382
60 - 64	2,223	459,365	1,190,217	0.042326	0.0048393	0.00020483
65 - 69	3,108	431,223	1,084,588	0.038570	0.0072074	0.00027799
70 - 74	3,866	374,246	834,024	0.029659	0.0103301	0.00030638
75 - 79	3,345	255,754	622,221	0.022127	0.0130790	0.00028940
80 - 84	2,719	167,495	382,303	0.013595	0.0162333	0.00022070
85 - 89	1,692	85,585	192,410	0.006842	0.0197698	0.00013527
90 +	906	42,224	95,467	0.003395	0.0214570	0.00007285
Sum	22,250	11,100,876	28,120,065	1.0		0.001877
Rate per 100,000					200.4346	187.7



Age adjusted rates

- What would the comparable death rate be in ONTARIO if it had Canada's age distributions?"

The comparable death rate in Ontario will be 187.7 deaths per 100000

Relative : SRR Standardized Rate Ratio

Absolute: SRD Standardized Rate Difference



Age/sex-adjusted rates [first method]

- Formula

- Age-specific expected events = $E_i = \left(\frac{e_i(m)}{p_i(m)} * P_i(m) \right) + \left(\frac{e_i(f)}{p_i(f)} * P_i(f) \right)$
- Adjusted rate = $\frac{\sum E_i}{\sum P_i} \times k$

Components of the formula

- $e_i(f)$ is the number of events for females in age group i .
- $e_i(m)$ is the number of events for males in age group i .
- $p_i(f)$ is the number of females in age group i the study population.
- $p_i(m)$ is the number of males in age group i the study population.
- $P_i(f)$ is the number of females in age group i in the Standard population.
- $P_i(m)$ is the number of males in age group i in the Standard population.



Age/sex-adjusted rates [Second method]

Formula

Components of the formula

E_i = Expected events

k = constant, multiple of 10

$$\text{Age/sex adjusted rate} = \sum E_i * k$$

$$E_i = [r_i(m) * W_i(m)] + [r_i(f) * W_i(f)]$$

$$W_i(f) = \text{Weight (females)}, W_i(f) = \frac{P_i(f)}{\sum P_i}, r_i(f) = \text{Rate(female)}, r_i(f) = \frac{e_i(f)}{p_i(f)}$$

$$W_i(m) = \text{Weight (males)}, W_i(m) = \frac{P_i(m)}{\sum P_i}, r_i(m) = \text{Rate(male)}, r_i(m) = \frac{e_i(m)}{p_i(m)}$$



Calculating an Age/Sex Adjusted Rate Using the Direct Method

	Study Population (Ontario)				Standard Population				METHOD 1	METHOD 2		
	deaths		population		(Canada 1991)		Crude Rate		Expected deaths	Population Weights		Expected deaths
	$e_{i(m)}$	$e_{i(f)}$	$e_{i(m)}$	$e_{i(f)}$	$P_{i(m)}$	$P_{i(f)}$	$r_{i(m)} = e_{i(m)}/p_{i(m)}$	$r_{i(f)} = e_{i(f)}/p_{i(f)}$	E_i	$P_{i(m)}/Total P$	$P_{i(f)}/Total P$	E_i
	Male	Female	Male	Female	Male	Female	Male	Female	AGE/SEX	Male	Female	AGE/SEX
AGE GRP												
<1 year	2	5	75,676	72,111	206,370	195,361	0.000026	0.000069	19.00	0.007339523	0.006947989	0.0000006757
1-4	17	13	311,955	296,313	793,902	757,536	0.000054	0.000044	76.50	0.028235023	0.026941671	0.0000027207
5-9	12	13	390,096	370,243	998,205	954,705	0.000031	0.000035	64.23	0.035501034	0.033953962	0.0000022843
10-14	14	9	379,922	360,157	980,454	932,534	0.000037	0.000025	59.43	0.034869722	0.033165453	0.0000021137
15 - 19	17	10	371,156	350,240	985,072	940,854	0.000046	0.000029	71.98	0.03503396	0.033461353	0.0000025600
20 - 24	32	15	378,228	368,943	1,067,654	1,041,341	0.000085	0.000041	132.67	0.037970978	0.03703516	0.0000047183
25 - 29	32	34	412,042	414,384	1,282,185	1,246,500	0.000078	0.000082	201.85	0.045600746	0.044331614	0.0000071788
30 - 34	60	62	500,645	494,192	1,312,036	1,285,944	0.000120	0.000125	318.57	0.046662393	0.045734435	0.0000113300
35 - 39	122	142	484,808	488,431	1,173,504	1,171,180	0.000252	0.000291	635.80	0.04173552	0.041652867	0.0000226122
40 - 44	174	269	425,652	439,665	1,077,008	1,061,763	0.000409	0.000612	1089.88	0.038303652	0.037761466	0.0000387615
45 - 49	307	424	391,670	398,599	844,091	830,034	0.000784	0.001064	1544.55	0.030019989	0.029520053	0.0000549316
50 - 54	494	528	301,936	306,702	673,195	666,661	0.001636	0.001722	2249.10	0.023942094	0.023709714	0.0000799891
55 - 59	787	736	247,139	254,079	618,181	620,200	0.003184	0.002897	3765.12	0.021985529	0.022057334	0.0001339059
60 - 64	1,252	971	225,081	234,284	578,610	611,562	0.005562	0.004145	5753.13	0.020578191	0.021750125	0.0002046093
65 - 69	1,799	1,309	205,114	226,109	497,864	586,692	0.008771	0.005789	7763.13	0.01770647	0.020865626	0.0002760949
70 - 74	2,161	1,705	164,342	209,904	364,284	469,730	0.013149	0.008123	8605.62	0.012955714	0.016705888	0.0003060579
75 - 79	1,813	1,532	105,385	150,369	255,603	366,627	0.017204	0.010188	8132.58	0.009090488	0.013039043	0.0002892343
80 - 84	1,455	1,264	62,541	104,954	142,234	240,076	0.023265	0.012043	6200.36	0.005058534	0.008538272	0.0002205151
85 - 89	777	915	26,909	58,676	62,193	130,221	0.028875	0.015594	3826.51	0.002211886	0.004631293	0.0001360893
90 +	347	559	9,976	32,248	25,490	69,976	0.034783	0.017334	2099.62	0.000906549	0.002488688	0.0000746728
Total	11,674	10,515	5,470,273	5,630,603	13,938,135	14,179,497	0.00213	0.001867	52609.65			0.0018710555
Rate per 100,000							213.41	186.75	187.106			187.106



PRACTICE TIME



Standardization

- Direct standardization
- **Indirect standardization**



Indirect standardization

- The indirect method is a mirror image of direct standardization. It involves applying the age-specific rates of the standard population to the study population to yield the expected deaths.
- Once these rates are obtained, the observed number of events in the study population is divided by the expected number of events to produce the **standardized mortality/morbidity ratio (SMR)**



Indirect standardization

- What would be expected number of events in the study population if its mortality experience (stratum-specific rates) was the same as that of the standard population?



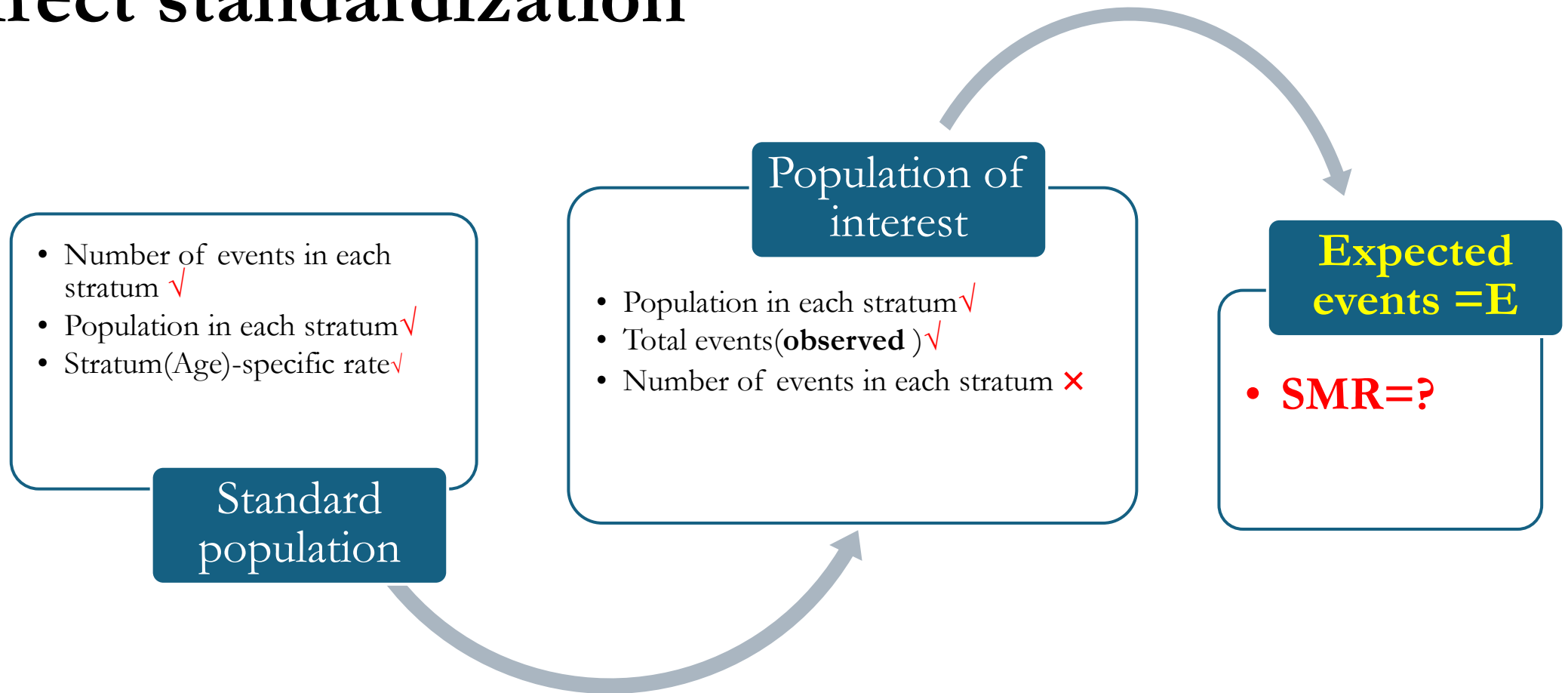
Indirect standardization

When to use indirect standardization?

- Stratum-specific rates are unavailable for the population of interest (i.e., the age-specific distribution of events in the study population is unknown)
- And/or when there are a small number of events in the study population (resulting in unstable age-specific rates).

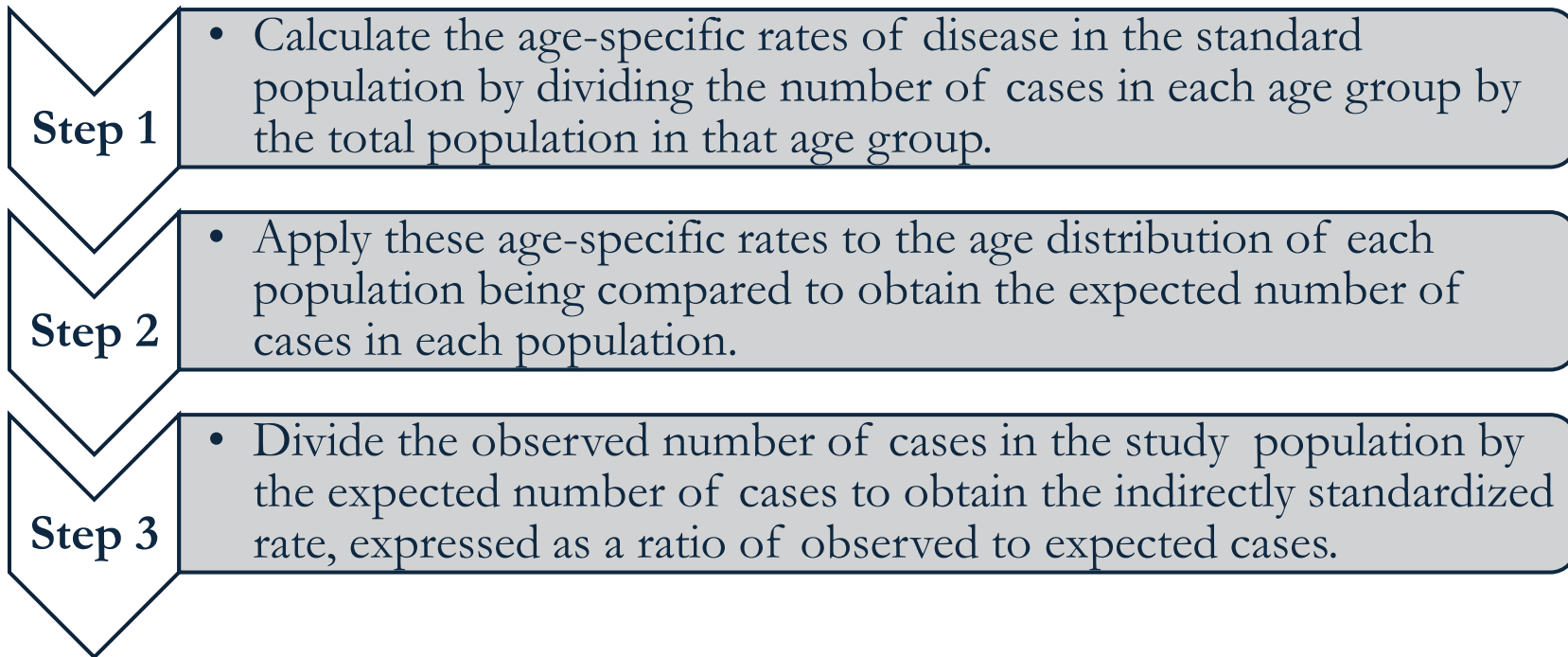


Indirect standardization





Steps to perform indirect standardization





Indirect standardization

Formula

$$SMR = \frac{\sum e}{\sum p_i \times R_i} * k$$

or

$$SMR = \frac{\text{Observed number of deaths (O)}}{\text{Expected number of deaths (E)}} * k$$

K=100

Components of the formula

e: is the total observed number of events in the study population.

p_i: is the population in age group **i** in the study population.

R_i: is the event rate in the standard population for the persons in age group **i**

$$E = R_i * p_i$$



Indirect Standardization: Calculating an SMR

	Standard Population (Ontario, 2001)			Study Population		
	Deaths	Population	Rate	Population	Deaths, Observed	Deaths, Expected
Age Groups	E_i	P_i	R_i	p_i	e	$E = R_i * p_i$
<1 year	50	147,787	0.00033832	951	-	0.322
1-4	30	608,268	0.00004932	4,433	-	0.219
5-9	25	760,339	0.00003288	6,141	-	0.202
10-14	28	740,079	0.00003783	6,369	-	0.241
15 – 19	40	721,396	0.00005545	6,241	-	0.346
20 – 24	47	747,171	0.00006290	5,088	-	0.320
25 – 29	66	826,426	0.00007986	5,128	-	0.410
30 – 34	122	994,837	0.00012263	5,705	-	0.700
35 – 39	264	973,239	0.00027126	6,934	-	1.881
40 – 44	443	865,317	0.00051195	6,776	-	3.469
45 – 49	731	790,269	0.00092500	5,940	-	5.495
50 – 54	1,022	608,638	0.00167916	5,541	-	9.304
55 – 59	1,523	501,218	0.00303860	4,526	-	13.753
60 – 64	2,223	459,365	0.00483929	3,666	-	17.741
65 – 69	3,108	431,223	0.00720741	3,009	-	21.687
70 – 74	3,866	374,246	0.01033010	2,837	-	29.307
75 – 79	3,345	255,754	0.01307897	2,431	-	31.795
80 – 84	2,719	167,495	0.01623332	1,658	-	26.915
85 – 89	1,692	85,585	0.01976982	983	-	19.434
90 +	906	42,224	0.02145699	381	-	8.175
Sum	22,250	11,100,876	0.00200435	84,738	203	192
SMR = Observed deaths / Expected deaths =						1.06



Indirect standardization

- $SMR = 100$, the risk of mortality is the same in both the expected and observed population..
- $SMR > 100$, the risk of mortality is higher in the observed population.
- $SMR < 100$, the risk of mortality is lower in the observed population.

Interpretation:

- There are 6% more events observed than expected
- The observed number of events in the study population exceeds the expected number by 6%, assuming the same event rate as that of the standard population.



Standardization

- Standardization allows for more accurate comparisons of disease rates between populations.
- Without standardization, differences in disease rates between populations may be attributed to the variable of interest (exposure, intervention) when they are actually due to differences in the distribution of confounding variables
- Standardization helps to isolate or remove the effect of confounding factors on disease rates.
- Standardization is crucial for ensuring the validity and reliability of epidemiological studies that compare disease rates across different populations or time periods



CHOICE OF THE STANDARD POPULATION

- Depends on the specific context and objective of the analysis.

Example:

In trying to summarize age-specific rates for geographic regions within a country, **the national population** could be used as a **standard**.

When examining the rates for a variety of countries, a **world population** or the **population for the appropriate continent** would be the suitable **standard**.



CHOICE OF THE STANDARD POPULATION

- It is important to use a standard population that is appropriate for the populations being compared and the health outcome of interest.
- The standard population should be representative of the populations being compared in terms of age distribution and other relevant demographic characteristics.
- The choice of the standard population should be clearly stated in the study methods and considered when interpreting the results.
- Common standard populations used in epidemiological studies include:
 - World Health Organization (WHO) World Standard Population
 - United States Standard Population (year 2000 or 2010)
 - European Standard Population



Comparison of SR(direct) and SMR's(Indirect)

- Rates that have been standardized by the direct method, using the same standard population, may be compared **in relative or absolute terms (i.e., as a ratio or as a difference)**.
 - We can obtain a "Standardized Rate Ratio" ("SRR") by dividing the (directly) standardized rate for country X by that of country Y.
- It is **"incorrect"** to compare indirectly adjusted rates for different areas.
 - The weights used in obtaining the indirectly standardized rates are the stratum sizes of the individual study populations rather than of a (common) standard population.



Adjusted rates (direct or indirect methods)

Pros:

- Both rates are statistically constructed based on a standard.
- Both methods are used for descriptive and analytical purposes [Epi].
- It removes the effect of the differences in population structure and allows for valid comparisons between our populations of interest.
- These rates can also be compared and ranked to the specific rates over time.

Cons:

- They can be inefficient when the number of categories of the confounding variable is relatively high (complex), and the sample size is relatively low or rare case events. (Direct)
- The choice of a standard/reference may vary depending on the nature of the information required.
- Reference rates may not be always comparable (Indirect).
- Indirect standardization cannot be used for time series analysis.



RECOMMENDATIONS

- Examine crude rates, age-specific rates, and counts before calculating adjusted rates.
- When there is little to no variation across age-specific rates or where there is no difference in the age structure of the populations, crude rates can be valid for comparisons.
- Consider direct standardization if there is at least 25 total deaths over all age groups.
- When fewer than 25 health events occurred over a time period, you may consider indirect age-adjustment.





What does the term 'adjusted rate' refer to ?

- A. A rate that applies only to the elderly population
- B. A rate modified to account for various confounding factors
- C. A rate calculated without consideration of any variables
- D. A rate that is always lower than the unadjusted rate
- E. None of the above





What is the purpose of standardization in epidemiology?

- A. Adjusting for population structure differences
- B. Comparing age distributions
- C. Calculating crude rates
- D. Estimating total population size
- E. None of the above





What is the key difference between crude rates and specific rates ?

- A. Specific rates are calculated using age distribution only
- B. Crude rates adjust for confounding factors
- C. Crude rates consider the total population, while specific rates focus on subsets
- D. Crude rates are more accurate than specific rates





- **How is the expected number of deaths calculated in direct standardization?**
 - A. Selecting a standard age group for analysis
 - B. Dividing the total expected deaths by the total population size
 - C. Summing up the observed deaths in each age group
 - D. Multiplying the rate by the standard population size

